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1. Deskripsi proyek

Saya membuat aplikasi pencatatan keuangan yang saya beri nama FinTrack. FinTrack adalah aplikasi pencatatan keuangan pribadi berbasis web. Aplikasi ini dapat memungkinkan pengguna untuk mencatat pemasukan dan pengeluaran, mengelola kategori transaksi, serta melihat laporan keuangan dalam beberapa periode waktu.

2. Daftar service

No.	Service	Port	Deskripsi	Database
1.	Auth Service	4001	Mengelola autentikasi dan manajemen user	MySQL
2.	Transaction Service	4002	Mengelola transaksi dan kategori	MariaDB
3.	Reports Service	4003	Membuat laporan keuangan	MariaDB
4.	Frontend	3000	User interface	-
5.	Nginx	80	API Gateway & Proxy	-
6.	Redis	6379	Token	-

3. Database

A. auth_service_db (MySQL) :

- Table : users

Kolom	Tipe data	Keterangan
Id	INT (PK)	Primary key, auto increment
name	VARCHAR(100)	Nama pengguna
email	VARCHAR(100)	Email (unique)
password	VARCHAR(255)	Password (hashed dengan bcrypt)
is_active	BOOLEAN	Status akun aktif
createdAt	TIMESTAMP	Waktu pembuatan
updatedAt	TIMESTAMP	Waktu update terakhir

B. catat_pengeluaran (MariaDB)

- Tabel : categories

Kolom	Tipe data	Keterangan
Id	INT (PK)	Primary key, auto increment
user_id	INT	Id pengguna pemilik kategori
name	VARCHAR(100)	Nama kategori
type	ENUM('income', 'expense')	Tipe kategori
is_default	BOOLEAN	Kategori bawaan sistem atau bukan
createdAt	TIMESTAMP	Waktu pembuatan

updatedAt	TIMESTAMP	Waktu update terakhir
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- Tabel : transactions

Kolom	Tipe data	Keterangan
Id	INT (PK)	Primary key, auto increment
user_id	INT	Id pengguna pemilik kategori
category_id	INT (FK)	Referensi ke tabel categories
type	ENUM('income', 'expense')	Tipe kategori
amount	DECIMAL(15,2)	Jumlah uang
notes	VARCHAR(500)	Catatan transaksi
transaction_date	DATE	Tanggal transaksi
createdAt	TIMESTAMP	Waktu pembuatan
updatedAt	TIMESTAMP	Waktu update terakhir

4. API Endpoint (Total 24 Endpoint)

A. Auth service endpoint

- Base url : /api/auth

Method	Endpoint	Deskripsi	Auth Required
POST	/api/auth/register	Registrasi pengguna baru	No
POST	/api/auth/login	Login pengguna	No
POST	/api/auth/refresh	Refresh access token	No
POST	/api/auth/logout	Logout pengguna	Yes

- Base url : /api/users

Method	Endpoint	Deskripsi	Auth Required
GET	/api/auth/profile	Mendapatkan profil pengguna	Yes
PUT	/api/auth/profile	Update profil pengguna	Yes
PUT	/api/auth/password	Ganti password	Yes
DELETE	/api/auth/account	Logout pengguna	Yes

B. Transaction service endpoint

- Base url : /api/categories

Method	Endpoint	Deskripsi	Auth Required
GET	/api/categories	Mendapatkan semua kategori	Yes
GET	/api/categories/:id	Mendapatkan kategori by ID	Yes
POST	/api/categories	Membuat kategori baru	Yes
POST	/api/categories/init	Inisialisasi kategori default	Yes
PUT	/api/categories/:id	Update kategori	Yes
DELETE	/api/categories/:id	Hapus kategori	Yes

- Base url : /api/transaction

Method	Endpoint	Deskripsi	Auth Required
GET	/api/transactions	Mendapatkan semua transaksi	Yes
GET	/api/transactions/:id	Mendapatkan transaksi by ID	Yes
POST	/api/transactions	Membuat transaksi baru	Yes
PUT	/api/transactions/:id	Update transaksi	Yes
DELETE	/api/transactions/:id	Hapus transaksi	Yes

C. Report service endpoint

- Base url : /api/reports

Method	Endpoint	Deskripsi	Auth Required
GET	/api/reports/daily	Laporan harian	Yes
GET	/api/reports/weekly	Laporan mingguan	Yes
GET	/api/reports/monthly	Laporan bulanan	Yes
GET	/api/reports/yearly	Update transaksi	Yes
GET	/api/reports/summary	Ringkasan keuangan	Yes

5. Fitur Aplikasi

A. Fitur Autentikasi

- Registrasi Pengguna
- Login/Logout
- Token Refresh
- Manajemen Profil
- Hapus Akun

B. Fitur Transaksi

- Catat Pemasukan
- Catat Pengeluaran
- Transaksi dikelompokkan berdasarkan kategori
- Kategori Custom
- Filter transaksi

C. Fitur Laporan

- Ringkasan transaksi per hari
- Ringkasan transaksi per minggu
- Ringkasan transaksi per bulan
- Ringkasan transaksi per tahun
- Ringkasan keseluruhan keuangan

6. Teknologi yang digunakan

A. Backend

Teknologi	Penggunaan
Node.js	Runtime environment
Express.js	Web framework
Sequelize	ORM untuk database
JWT	Autentikasi
bcryptjs	Password hashing
express-validator	Validasi input
express-rate-limit	Rate limiting
Helmet	Security headers
CORS	Cross-Origin Resource Sharing

B. Frontend

Teknologi	Penggunaan
Next.js	React framework
React	UI library
TypeScript	Type-safe JavaScript
TailwindCSS	CSS framework
bcryptjs	Password hashing
Framer Motion	Animasi
Chart.js	Visualisasi data
Axios	HTTP client

C. Database & cache

Teknologi	Penggunaan
MySQL	Database auth service
MariaDB	Database transaction & reports
Redis	Token & caching

D. Infrastruktur

Teknologi	Penggunaan
Docker	Untuk container
Docker Compose	Mengelola banyak container
Nginx	API Gateway & Proxy

7. Keamanan

A. Fitur keamanan

Fitur	Deskripsi
Password Hashing	Bcrypt dengan salt rounds 12
JWT Token	Access token (15m) + Refresh token (7d)
Token Blacklisting	Revoke token saat logout menggunakan Redis
Rate Limiting	Pembatasan request untuk mencegah brute force
Input Validation	Validasi semua input menggunakan express-validator
Helmet.js	Security headers otomatis
CORS	Pembatasan origin yang diizinkan

SQL Injection Prevention	Menggunakan ORM (Sequelize)
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B. Rate limit

Endpoint Type	Limit	Window
General API	100 requests	15 menit
Auth (login/register)	10 requests	15 menit
Password Change	5 requests	1 jam

8. Cara menjalankan proyek

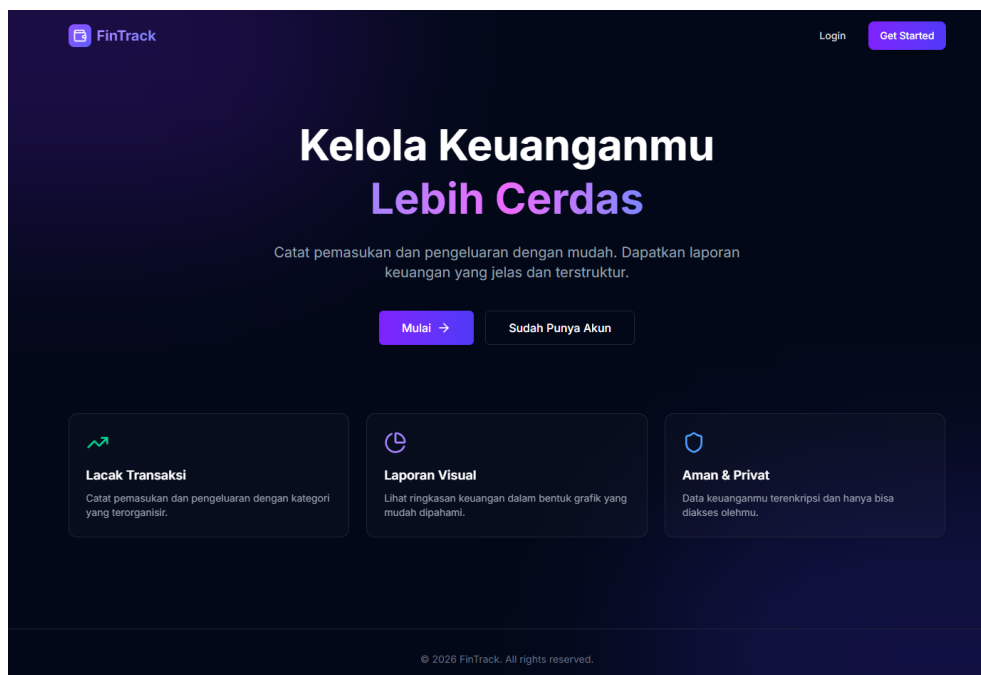
Menjalankan proyek bisa dengan melakukan clone dari link berikut :

<https://github.com/justajay224/FinTrack.git>

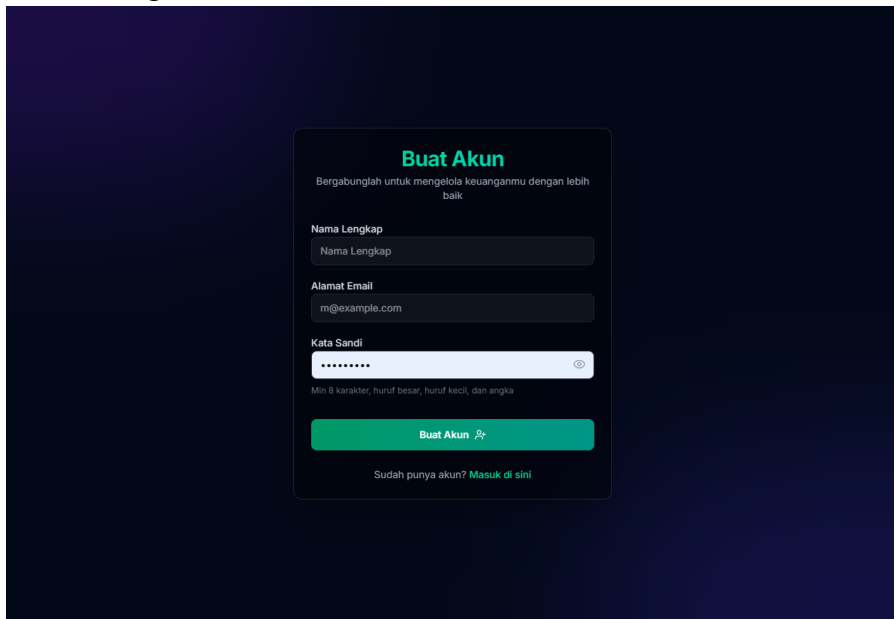
Kemudian untuk caranya bisa dilihat pada file README.md. Disana sudah diterangkan Langkah-langkah inisialisasi proyek.

9. Tampilan UI

A. Halaman utama

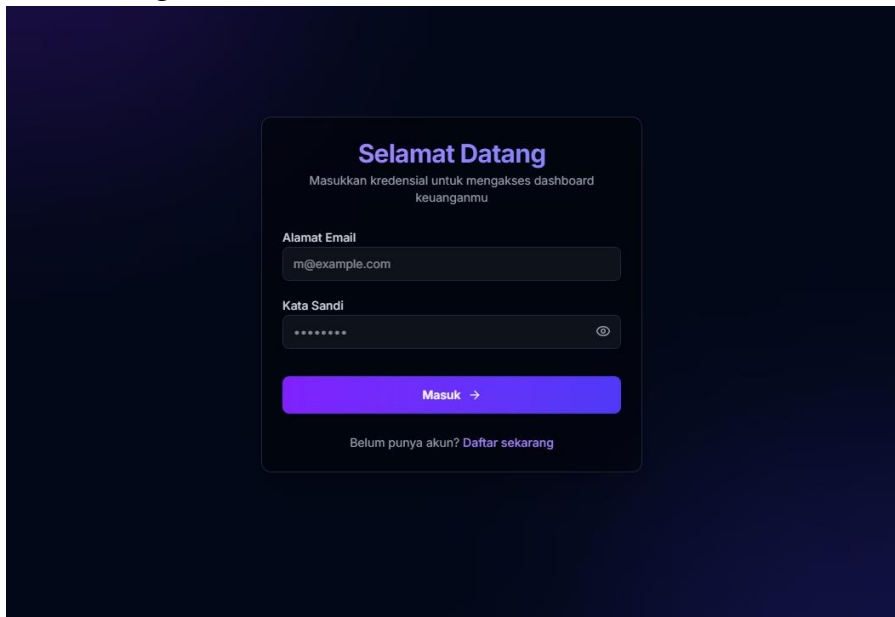


B. Halaman register



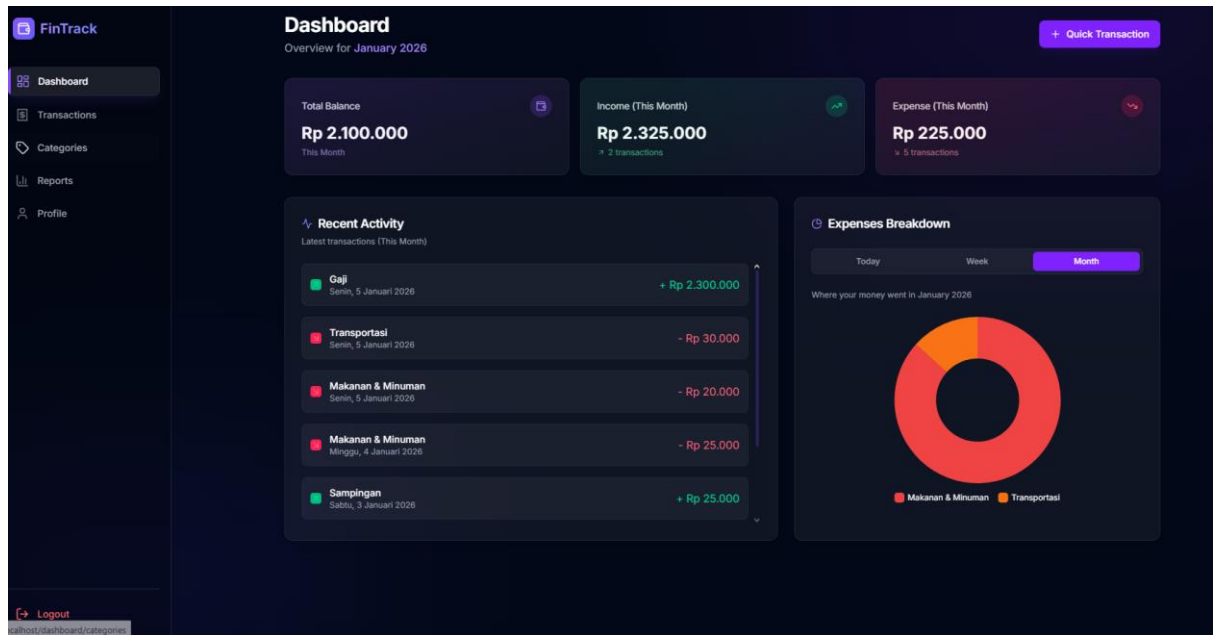
The screenshot shows a registration form titled "Buat Akun" (Create Account) in teal. Below the title is a subtitle: "Bergabunglah untuk mengelola keuanganmu dengan lebih baik" (Join to manage your finances better). The form contains three input fields: "Nama Lengkap" (Full Name) with the placeholder "Nama Lengkap", "Alamat Email" with the placeholder "m@example.com", and "Kata Sandi" (Password) with a masked password "*****" and a toggle icon. Below the password field is a note: "Min 8 karakter, huruf besar, huruf kecil, dan angka" (Minimum 8 characters, uppercase, lowercase, and numbers). At the bottom of the form is a teal button labeled "Buat Akun" with a right arrow, and a link "Sudah punya akun? Masuk di sini" (Already have an account? Log in here).

C. Halaman login

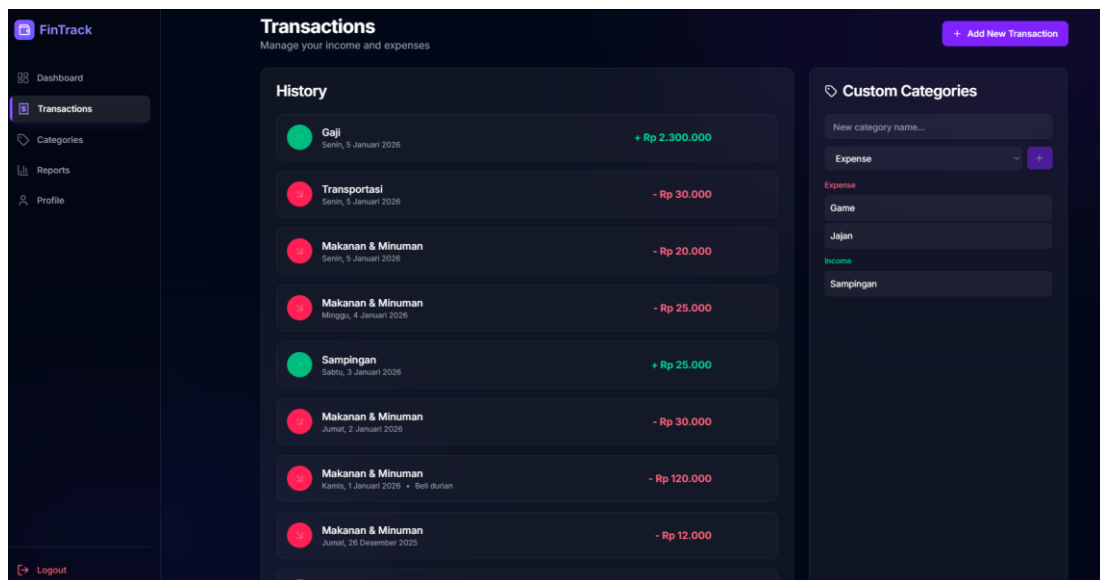


The screenshot shows a login form titled "Selamat Datang" (Welcome) in purple. Below the title is a subtitle: "Masukkan kredensial untuk mengakses dashboard keuanganmu" (Enter credentials to access your finance dashboard). The form contains two input fields: "Alamat Email" with the placeholder "m@example.com" and "Kata Sandi" (Password) with a masked password "*****" and a toggle icon. At the bottom of the form is a purple button labeled "Masuk" with a right arrow, and a link "Belum punya akun? Daftar sekarang" (Don't have an account? Sign up now).

D. Halaman dashboard user



E. Halaman transaksi



F. Pop up membuat transaksi baru

The 'Add Transaction' pop-up form is displayed over a blurred background of transaction history. The form has a dark theme with a title bar 'Add Transaction' and a close button. It contains two tabs: 'Expense' (active) and 'Income'. The 'Amount' field is set to 'Rp 0'. The 'Category' field has a dropdown menu labeled 'Select Category'. The 'Date' field is set to '05/01/2026' with a calendar icon. The 'Notes' field is labeled 'Optional'. At the bottom, there are 'Cancel' and 'Save Transaction' buttons.

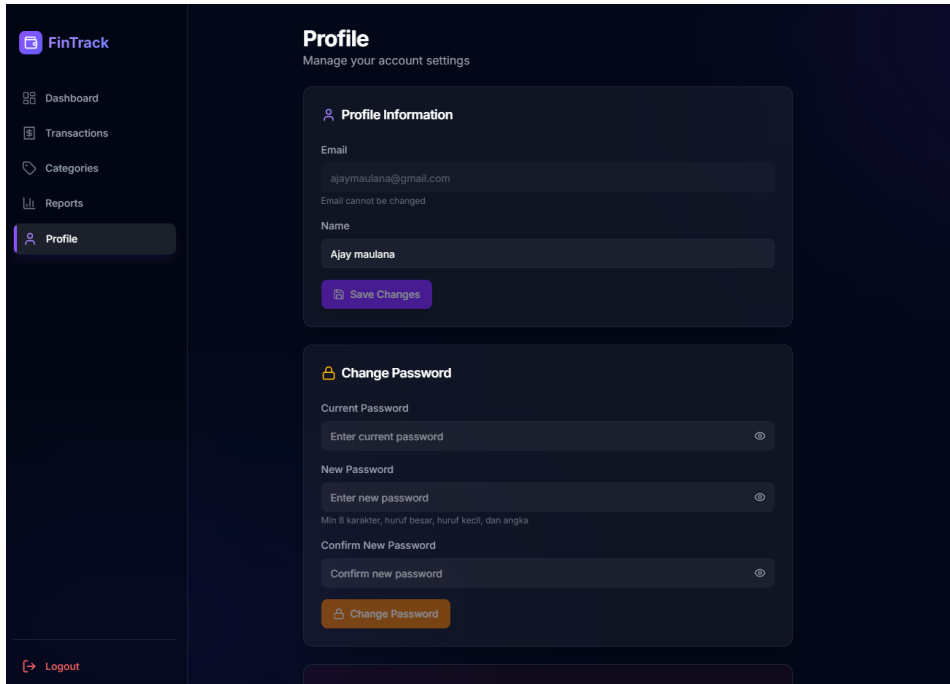
G. Halaman manajemen kategori

The 'Categories' page is titled 'Categories' with the subtitle 'Manage your income and expense categories'. It has two tabs: 'Expense' (active) and 'Income'. The 'Add New Category' section has a text input field 'Enter category name...' and an '+ Add Category' button. Below this is a 'Custom Categories' section with a list of categories: 'Game' and 'Jajan'. The 'Default Categories' section (marked 'Cannot be deleted') contains a grid of categories: 'Belanja', 'Hiburan', 'Kesehatan', 'Lainnya', 'Makanan & Minuman', 'Olahraga', 'Pakaian', 'Pendidikan', 'Rumah Tangga', and 'Tagihan & Utilitas'. A 'Transportasi' category is also listed below the grid. A 'Logout' button is in the bottom left corner.

H. Halaman Report



I. Halaman profile



The screenshot shows the 'Profile' page of the 'FinTrack' application. The left sidebar contains navigation links: Dashboard, Transactions, Categories, Reports, and Profile (which is highlighted). The main content area is titled 'Profile' with the subtitle 'Manage your account settings'. It features two sections: 'Profile Information' and 'Change Password'. The 'Profile Information' section has input fields for 'Email' (containing 'ajaymaulana@gmail.com' with a note 'Email cannot be changed') and 'Name' (containing 'Ajay maulana'), followed by a 'Save Changes' button. The 'Change Password' section has input fields for 'Current Password', 'New Password' (with a note 'Min 8 karakter, huruf besar, huruf kecil, dan angka'), and 'Confirm New Password', followed by a 'Change Password' button. A 'Logout' link is located at the bottom left of the sidebar.

FinTrack

- Dashboard
- Transactions
- Categories
- Reports
- Profile**

Profile
Manage your account settings

Profile Information

Email
ajaymaulana@gmail.com
Email cannot be changed

Name
Ajay maulana

Save Changes

Change Password

Current Password
Enter current password

New Password
Enter new password
Min 8 karakter, huruf besar, huruf kecil, dan angka

Confirm New Password
Confirm new password

Change Password

Logout